WEEK 7 – DATA SCIENCE AND VISUALIZATION

Feature Selection, PCA and Explained Variance

Project	Dataset	Syllabus Mapping
Data Analysis Project	Cleaned_Dataset.csv	Unit 4

1. Introduction

Week 7 explores dimensionality reduction. Feature selection ranks and retains useful original columns, while Principal Component Analysis (PCA) transforms numerical features into fewer components that preserve as much variation as possible. The target variable is kept separate from the input features.

2. Objectives

• Loadand inspectthe cleaned dataset.
• Separate numerical input features from the target variable when available. •
Impute missing values and standardize numerical features.
• Rank features using mutual information when a target is available.
• Apply PCA and inspect explained variance.
• Plot cumulative explained variance and the first two components.
• Save the reduced dataset separately.

3. Dataset Used

Dataset: **Cleaned_Dataset.csv**. The Colab preview showed 156,944 rows and 53 columns. The visible columns included **Late_delivery_risk**, which is used as the target for supervised feature selection if present. The code checks whether the column exists.

4. Tools and Technologies

Python, Pandas, NumPy, Matplotlib and Scikit-learn.

5. Load Dataset and Inspect

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv('/content/Cleaned_Dataset.csv')
print("Dataset shape:", df.shape)
print("Columns:", df.columns.tolist())
display(df.head())
```

6. Define Target and Numerical Inputs

Thetargetcolumn mustnot beincluded amongthe inputfeatures. If Late_delivery_risk is missing, PCA can still be demonstrated on numerical columns, but supervised feature selection will be skipped.

```
target = 'Late_delivery_risk'
if target in df.columns:
    work = df.dropna(subset=[target]).copy()
    y = work[target]
    X = work.select_dtypes(include=np.number).drop(columns=[target],
    errors='ignore')
    print("Target:", target)
    print("Target classes:", y.nunique())
else:
    work = df.copy()
    X = work.select_dtypes(include=np.number)
    y = None
    print("Target not found; supervised selection will be
    skipped.")
print("Numerical input shape:",
X.shape)
```

7. Impute and Scale Numerical Data

Missingnumericalvalues are filledwiththe median, and StandardScaler puts features on comparable scales before PCA. A reproducible sample of up to 20,000 rows keeps the demonstration manageable.

```
from sklearn.impute import
SimpleImputer
from sklearn.preprocessing import
StandardScaler
max_rows = min(20000, len(X))
X_sample = X.sample(n=max_rows, random_state=42) if len(X) >
max_rows else X.copy() y_sample = y.loc[X_sample.index] if y is
not None else None
X_imputed =
SimpleImputer(strategy='median').fit_transform(X_sam
ple) X_scaled =
StandardScaler().fit_transform(X_imputed)
print("Rows used:", X_scaled.shape[0])
print("Numerical features used:",
X_scaled.shape[1])
```

8. Feature Selection with Mutual Information

Mutual information estimateshow informative each inputfeatureisabout the target. SelectKBest keeps the top features. This is run only when the target exists and has at least two classes.

```
from sklearn.feature_selection import SelectKBest, mutual_info_classif

if y_sample is not None and y_sample.nunique() >= 2 and X_sample.shape[1] > 0:
    X_for_selection = pd.DataFrame(
        X_imputed, columns=X_sample.columns, index=X_sample.index
    )
    k = min(10, X_for_selection.shape[1])
    selector = SelectKBest(score_func=mutual_info_classif, k=k)
    selector.fit(X_for_selection, y_sample)
    scores = pd.Series(selector.scores_, index=X_for_selection.columns)
    scores = scores.sort_values(ascending=False)
    print(f"Top {k} features:")
    display(scores.head(k).to_frame('Mutual Information Score'))
else:
    print("Supervised feature selection skipped: target/features unavailable.")
```

9. Principal Component Analysis (PCA)

PCA forms newcomponents fromscaled numericalinputs.n_components=0.95 requests enough components to explain at least 95% of variance, where achievable. The retained component count depends on the data.

```
from sklearn.decomposition import PCA

if X_scaled.shape[1] >= 1:
    pca = PCA(n_components=0.95, random_state=42)
    X_pca = pca.fit_transform(X_scaled)
    print("Original numerical shape:", X_scaled.shape)
    print("PCA-transformed shape:", X_pca.shape)
    print("Components retained:", pca.n_components_)
    print("Explained variance retained:",
          round(pca.explained_variance_ratio_.sum(), 4))
else:
    X_pca = np.empty((len(X_scaled), 0))
    print("PCA skipped: no numerical input features.")
```

10. Cumulative Explained Variance Plot

```
if X_scaled.shape[1] >= 1:
    pca_all = PCA()
    pca_all.fit(X_scaled)
    cumulative = np.cumsum(pca_all.explained_variance_ratio_)

    plt.figure(figsize=(8, 4))
    plt.plot(range(1, len(cumulative) + 1), cumulative, marker='o')
    plt.axhline(0.95, linestyle='--', label='95% variance')
    plt.xlabel('Number of Principal Components')
    plt.ylabel('Cumulative Explained Variance')
    plt.title('PCA: Cumulative Explained Variance')
    plt.legend()
    plt.grid(True, alpha=0.3)
    plt.tight_layout()
    plt.show()
```

11. Plot First Two Principal Components

Atwo-dimensionalscatter plotprovides acompact viewofthe dataset. If a target exists, points are colored by the target label. This plot is exploratory and does not by itself prove accurate classification.

```
if X_scaled.shape[1] >= 2:
    pca_2d = PCA(n_components=2, random_state=42)
    X_2d = pca_2d.fit_transform(X_scaled)
    plt.figure(figsize=(8, 5))

    ify_sampleis not None:
        labels =y_sample.astype(str)
        for label in labels.unique()[:10]:
            mask=  (labels == label).to_numpy()
             plt.scatter(X_2d[mask, 0], X_2d[mask, 1],
                         s=10, alpha=0.5, label=label)
        plt.legend(title=target, bbox_to_anchor=(1.02, 1), loc='upper left')
    else:
        plt.scatter(X_2d[:, 0], X_2d[:, 1], s=10, alpha=0.5)

    plt.xlabel('Principal     Component     1')
    plt.ylabel('Principal     Component     2')
    plt.title('First Two Principal Components')
    plt.tight_layout()
    plt.show()
else:
    print("Two-component plot skipped: fewer than two numerical features.")
```

12. Save the PCA-Reduced Dataset

```
  if
  X_pca.shape[1] >
  0:
     pca_cols = [f'PC{i+1}' for i in
     range(X_pca.shape[1])]
     reduced_df = pd.DataFrame(X_pca,
     columns=pca_cols)
     if y_sample is not None:
reduced_df[target] =
y_sample.reset_index(drop=True)
     reduced_df.to_csv('Week7_pca_reduced_dataset.csv
     ', index=False)
     print("Saved: Week7_pca_reduced_dataset.csv")
     print("Saved shape:", reduced_df.shape)
```

13. Results to Record

Run the notebookandcopyactualresults into the table. Feature rankings, component counts and variance values depend on the data and must not be guessed.

Item	Record from output
Dataset dimensions	Printed shape
Rows used for PCA	Output from Section 7
Numerical features used	Output from Section 7
Top selected features	Top feature names and scores from Section 8
PCA components retained	Output from Section 9
Explained variance retained	Output from Section 9
Reduced dataset dimensions	Output from Section 12

14. Interpretation

Feature selection retainsoriginal columns ranked as useful for predicting the target. PCA creates new components that combine the original numerical features and summarize their variation. The cumulative variance plot helps assess how many components retain a chosen share of information; the two-component plot gives a simplified visual overview.

15. Conclusion

Week7 demonstratesdimensionality reduction using mutual-information feature selection and PCA. Numerical values are imputed and standardized before PCA. The transformed data is saved in a separate file so the cleaned source dataset remains unchanged. Final conclusions must be based on the notebook outputs.

16. Project Status

Week7:DimensionalityReduction — notebook steps prepared; run the cells and record the observed results before submission.